MOZAMMAL HOSSAIN

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EDUCATION

BSc in Computer Science and Engineering
Shahjalal University of Science and Technology (Sylhet, Bangladesh) February 2018 - March 2023
CGPA - 3.60 out of 4.0 (Having a CGPA of **3.83** for the last four consecutive semester)

WORK EXPERIENE

IQVIA, Durham, North Carolina, USA
A Fortune 500 Global Leader in Healthcare Analytics and Technology
Software Development Engineer I (Remote) March 2024 - Present
Received IQVIA Impact Award for Outstanding Performance in the AI Assistant Team

- Contribute to the AI Assistant team within the Orchestrated Analytics branch, focusing on AI-driven solutions delivering real-time insights for the life sciences industry.
- Actively contributing to the entire AI workflow for IQVIA's next-generation data exploration and orchestration agents for the pharmaceutical industry, delivering real-time insights.
- Design and develop sophisticated multi-agent AI systems, ensuring modular and scalable architectures.
- Implement and optimize advanced techniques to enhance agents' reasoning capabilities and response time.
- Engineer advanced tool-calling mechanism, including MCP and sophisticated context engineering for improving the agent's performance.
- Currently integrating the A2A protocol into our agent layer for more convenient orchestration.
- Fine-tuned LLMs using advanced fine-tuning techniques for pharmaceutical use cases (LoRA, QLoRA).
- Collaborate on large-scale pharmaceutical datasets(over multi-petabytes of data), strengthening expertise in scalable AI solutions for healthcare applications.

Dviz Technologies, Delaware, USA
Software Engineer Data & AI (Remote) July 2023 - February 2024

- Contributed to the development of backend solutions for RAG applications with fine-tuned LLMs.
- Built and optimized two AI-powered MVPs for data-driven decision-making platforms.
- Designed a statistical model and performed time-series data analysis for a fintech solution, enabling predictive analytics and actionable insights.

RESEARCHES AND PUBLICATIONS

DEEP LEARNING IN PROSTATE CANCER DIAGNOSIS AND GLEASON GRADING IN HISTOPATHOLOGY IMAGES: AN EXTENSIVE STUDY(Published)

- In this paper, I explored different aspects of applying Deep Learning for diagnosing prostate cancer and Gleason grading in histopathology images.
- Explored image pre-processing, post-processing, and evaluation techniques and their limitations, as well as analyzed existing deep learning methodological approaches and their impact on differentiating various prostate cancer Gleason grades
- Identified promising solutions to resolve obstacles faced in prostate cancer detection and Gleason grading, and discussed existing limitations and future scope in the field

EXPLORING SPATIAL AND TEMPORAL-AWARENESS SOLUTIONS IN SYNCHRONIZED, REMOTE COLLABORATIVE AUGMENTED REALITY(THESIS)

- Developed an AR system for exploring spatial and temporal-awareness solutions in synchronized, remote collaborative augmented reality.
- Designed a study with two design factors: representation of the user (mini avatar and large avatar + Ray) and update style (dynamic and static)
- Collected user reviews and selection time to analyze the results
- Found that participants preferred the dynamic update style and large avatar representation for exploring spatial and temporal-awareness solutions in AR
- This research was supervised by Khalad Hasan(Assistant Professor, UBC).

INTERPRETING THE LATENT SPACE OF CONVOLUTIONAL VARIATIONAL AUTOENCODERS FOR PERSON-AGNOSTIC MUSCLE-MOVEMENT ARTIFACT DETECTION IN EEG SIGNALS (IN PROGRESS)

- Currently working on developing a Convolutional Variational Autoencoder (CVAE) for muscle-movement artifact detection in EEG signals.
- Actively implementing preprocessing steps, including EEG topographic map generation and latent space disentanglement techniques to separate artifact-related components.
- Future work includes reconstructing clean EEG signals using the disentangled latent space components and evaluating model performance using metrics like SSIM, MSE, and SNR.
- Explainable AI (XAI) techniques, such as attention visualization and latent space exploration, will be integrated to interpret the model's artifact detection capabilities.
- The model will be benchmarked against traditional (ICA) and modern deep learning methods to demonstrate improvements in generalization and interpretability.

PROJECTS

FrameFlow(Link)

- Achieved Winner status in the **Nano Banana 48-hour Hackathon** by designing and deploying FrameFlow.
- Developed a **dynamic prompt generation** pipeline where Gemini (Text) first brainstorms cinematic scenes based on a user's theme to generate customised input for image creation.
- Built a **context-aware workflow** by chaining API calls to Gemini (Image Preview) to maintain thematic consistency and sustain user facial identity across all generated images.
- Introduced a **multi-modal album design** feature where Gemini (Text) performs visual analysis on generated images to craft fully customised layouts, fonts, and colour palettes.
- Implemented effective context preservation throughout the entire AI pipeline, transforming a simple user theme into a complete, professional visual narrative.
- Delivered a smooth user experience that simplifies the complex multi-model chain into an intuitive and engaging **"theme-to-album"** creation process.

FOCUS

- This is a full-stack community-based web app. It connects current undergraduate students with their alumni.
- This app has **three sections: job resources & QA, studentshub, and higher education & resources**.
- Built real-time messaging and notification system using **Socket.io**.
- Integrated **Cloudinary** as a **CDN** to serve / store media content fast and reduce loading time also used **presigned** URLs to control access.
- Implemented **JWT** based authentication and authorization system from scratch.

- **Tools/Tech: Next.js, Express.js, Socket.io, Mongodb, Sendgrid**

RHYTHMICSHARE

- A music-sharing platform where users may listen to music, share music, and write music reviews. They can also like and comment on specific music.
- Leveraged **Graphene** to make **Query/Mutation resolvers** and exposed GraphQL API endpoint and used **Apollo client** to consume the GraphQL API.
- Utilized **Django ORM** for seamless database integration and employed **PostgreSQL** to store and manage the platform's data effectively.
- Designed the UI with **Material UI** and implemented JWT for secure user access.
- **Tools/Tech: Django, React, PostgreSQL, GraphQL, Graphene, Apollo**.

SKILLS

Language	Python, JavaScript, TypeScript, Java, C, C++, Bash, SQL
Tools/Tech	FastAPI, Django, DRF, Celery, Flask, Node, Express, Next, React, GraphQL
Database	MySQL, PostgreSQL, MongoDB, VectorDBs, Knowledge Graph(Neo4j)
AI Tools/Tech	LangGraph, LangChain, LangSmith, LlamaIndex, MCP, A2A, vLLM, n8n, ADK, Vertex AI
ML Tools/Tech	PyTorch, Keras, TensorFlow, Scikit-Learn, XGBoost, Matplotlib, NumPy, Pandas
Version Control	Git, GitHub, GitLab
Data Tools/Tech	ETL, ELT, Scrapy, PySpark, Airflow, Kafka

EXTRA-CURRICULAR ACTIVITIES

- Conducted several workshops on **Machine Learning and Deep Learning**
- **2nd runner-up** at 2017 Bangladesh Physics Olympiad Chittagong Region
- Participated in the 2017 Bangladesh Physics Olympiad at the National Level
- Former Cricket Player at NCA Chittagong